PIYUSH SINGH

Associate Software Developer | Software Development Engineer | Full-Stack & AI/ML Developer
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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI) student focused on software development, full-stack engineering and AI/ML applications. Hands-on experience with Python, TypeScript, JavaScript, MERN, .NET, React Native, FastAPI, Django, REST APIs, SQL/SQLAlchemy, MongoDB, machine learning, Docker and Git. Built software projects involving authentication, data processing, dashboards, API design, AI integration, ML workflows and deployment-oriented architectures.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Full-Stack Development: MERN (MongoDB, Express.js, React, Node.js), .NET, FastAPI, Django, Django REST Framework, REST APIs
Frontend & Mobile: React, React Native, Vite, HTML/CSS, JavaScript, Streamlit
AI / Machine Learning: Machine Learning, Scikit-learn, Pandas, NumPy, Generative AI, Recommendation Systems
Databases: MongoDB, SQLAlchemy, PostgreSQL-ready architecture, SQLite
Software Engineering: JWT Authentication, Docker, Docker Compose, Git, GitHub
Visualization: Plotly, Matplotlib, Chart.js, Three.js

PROJECTS

AURA Finance Controller | *React, FastAPI, Python, Pandas, Docker, Three.js, AI APIs*
- Built an AI-assisted financial reconciliation platform with a React frontend and FastAPI/Python backend for data ingestion, normalization, reconciliation, exception detection and audit export.
- Designed a deterministic reconciliation engine and isolated AI to exception investigation, demonstrating separation of concerns, explainable workflows and controlled AI integration.
- Added Docker/Docker Compose support, CSV audit export, interactive 3D ledger visualization and failure-simulation workflows.

AI Learning Platform — Regional Languages | *React, Vite, FastAPI, SQLAlchemy, JWT, PostgreSQL/SQLite, Python*
- Developed a full-stack AI learning platform with React/Vite frontend and FastAPI backend, including authentication, protected routes, file processing, analytics and admin workflows.
- Implemented PDF/DOCX/PPTX/image document extraction, multilingual translation, AI summarization, question generation, chatbot functionality and voice assistance.
- Structured the backend into routes, services, models, schemas and utilities, with PostgreSQL-ready SQLAlchemy models and Render/Vercel deployment architecture.

AutoML Studio | *Python, Streamlit, Pandas, NumPy, Scikit-learn, XGBoost, Plotly*
- Built an end-to-end AutoML web application that accepts CSV datasets, profiles data, detects classification versus regression tasks and applies reusable preprocessing pipelines.
- Implemented imputation, one-hot encoding, scaling and model comparison across multiple algorithms with cross-validation, training/prediction timing and evaluation metrics.
- Added model leaderboard, interactive diagnostics, cleaned-data export and downloadable trained-model bundles with preprocessing metadata.

AI-Based Project Management System — Tech Expo 2026 | *React, Express.js, MongoDB Atlas, JWT*
- Built an AI-based project management system using React, Express.js, MongoDB Atlas and JWT authentication; secured 1st Position at Tech Expo 2026.

EDUCATION

B.Tech in Computer Science Engineering (Artificial Intelligence) — Babu Banarasi Das University | *Expected Graduation: 2027*
Relevant Coursework: DBMS, Machine Learning, Statistics, SQL, Data Analytics, Data Visualization

ACHIEVEMENTS

- 1st Position — Tech Expo 2026: AI-Based Project Management System using React, Express.js, MongoDB Atlas and JWT.
- Active GitHub portfolio with full-stack, AI/ML and software engineering projects.